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Rock Physics Insights for Unconventional Shale Reservoirs: A Midland Basin Case Study

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Abstract

Unconventional reservoirs exhibit significant heterogeneity and complex mineralogy. Variations in production from horizontal wells underscore the challenges posed by these source rocks. Seismic inversion is a crucial tool for understanding spatial variability. By integrating rock physics models with seismic results, we can identify lateral variations in elastic lithofacies, mineral composition, kerogen content, and porosity. Additionally, reservoir pressure estimation is vital for successful production. In this study, we present lithoclassification results and predictions of reservoir properties and pore pressure derived from prestack Amplitude-Variation-with Angle (AVA) inversions in the Midland Basin, West Texas.

Despite significant lateral variation in unconventional reservoirs, mature kerogen-rich shales hold substantial hydrocarbon reserves. Identifying optimal locations for horizontal wells—based on reservoir and completion quality—is crucial for maximizing production. Reservoir Quality (RQ), determined by mineralogy, porosity, and kerogen content, defines reservoir potential. Completion Quality (CQ), influenced by factors like pore pressure, ensures effective hydraulic fracturing. An illustrative example is the Sprayberry and Wolfcamp formation in the Permian Basin, West Texas. In this case study, prestack AVA inversion was performed from Clearfork to Atoka formations in the Midland basin. Acoustic impedance and V_p/V_s ratio guided litho-classification. Petrophysical interpretation, combined with elastic properties, generated a probability density function (PDF) from well data. Applying PDFs to Seismic AVA inversion yielded lithological descriptions and associated probabilities. Additionally, a rock physics model, incorporating effective stress estimation, predicted reservoir properties (e.g., clay volume, carbonate content, kerogen, porosity, and pore pressure) based on AVA inversion results and lithological information.

Introduction

Mature kerogen-rich shales are known to contain substantial amounts of hydrocarbons, although production from wells in such unconventional reservoirs often exhibits significant lateral variation. To optimize production, it is crucial to identify locations for horizontal wells that possess both good reservoir and completion quality. Reservoir Quality (RQ) encompasses various properties such as mineralogy, porosity, and kerogen content, which collectively define the reservoir's potential. Completion Quality (CQ), on the

other hand, includes properties like pore pressure that determine the potential for creating and maintaining a large surface area in contact with the reservoir through hydraulic fracturing (Suarez-Rivera et al., 2011).

In this case study, pre-stack AVA inversion was carried out from Clearfork to Atoka formations of the Midland basin. Estimated acoustic impedance and V_p/V_s ratio were used for litho-classification. In this process petrophysical interpretation was integrated with measured elastic properties to generate the probability density function (PDF) for the mentioned stratigraphic sequence obtained from well. By integrating petrophysical, geological, and geomechanical interpretations from various wells with elastic properties derived from the seismic data, we were able to estimate reservoir properties and geomechanical behavior across the entire study area. The RQ and CQ were then determined from these integrated, reservoir and geomechanical models, to identify the sweet spots and inform well design.

Similarly, mature kerogen-rich shales in other regions, such as the Sprayberry and Wolfcamp formations of the Permian Basin in West Texas, also contain large amounts of hydrocarbons despite considerable lateral variation in production (Glaser et al., 2013). The Midland Basin, the eastern subbasin of the Permian Basin, currently targets the Sprayberry and Wolfcamp formations using horizontal drilling. In this case study, pre-stack AVA inversion was performed from the Clearfork to Atoka formations of the Midland Basin. The estimated acoustic impedance and V_p/V_s ratio were used for litho-classification. Petrophysical interpretation was integrated with measured elastic properties to generate probability density functions (PDFs) for the stratigraphic sequence obtained from well data. These PDFs were applied to Seismic AVA inversion to generate lithological descriptions and associated probabilities. A rock physics model, developed using the approach described by Sayers and Dasgupta (2019), included effective stress estimation. This model was used to predict reservoir properties such as the volume of clay, carbonate, kerogen, porosity, and pore pressure from AVA inversion results and lithological descriptions.

A methodology has been developed to generate a mineralogy-based lithofacies log column for organic mudstones. This technique uses the dry-weight mineralogy output from geochemical tools in addition to other logs (resistivity, neutron porosity, etc.) to generate a log which defines the different organic mudstone classes that exist over the interval of interest. The identification of conventional carbonate stringers is also possible using this methodology. These thin beds are important for parasequence definition.

Input Data

The log data used in this paper are from a vertical well in the Clearfork to Atoka formations of the Midland Basin in which various types of open-hole logging data were acquired (S Dasgupta, 2022). This well has undergone various types of open-hole logging data acquisition. Figure 1 displays the depth-wise variation of several parameters, including lithology, P-impedance, S-impedance, bulk density, lithofacies, overburden stress, effective stress, and pore pressure for the well under consideration.

The petrophysical analysis in this study is thorough and utilizes multiple logging techniques. These techniques include elemental spectroscopy, nuclear magnetic resonance (NMR), neutron, density, and resistivity logs. Mineralogical composition and kerogen content are estimated using inelastic and capture spectroscopy measurements, while porosity and fluid saturation are determined by integrating data from NMR, neutron, density, and resistivity logs.

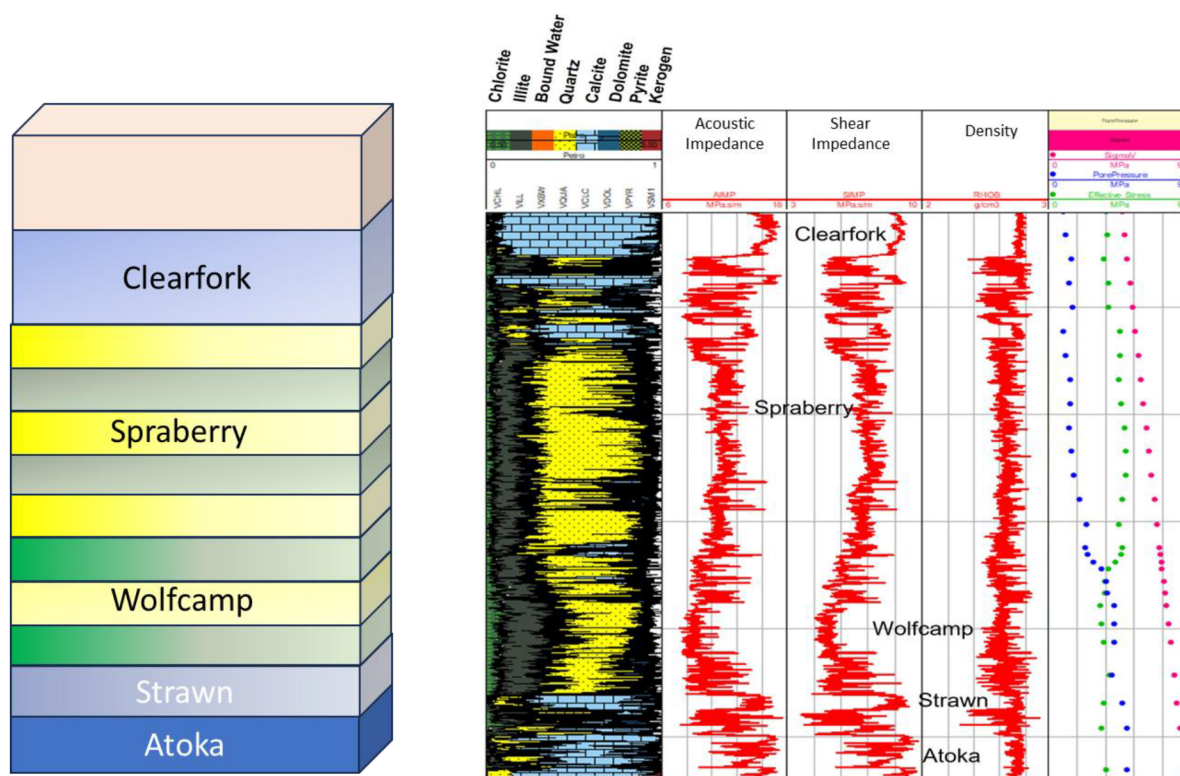


Figure 1—The Litho-column of the formations along the accurate Petrophysical evaluation results (Volumes of Chlorite, Illite, Bound Water, Qtz, Calcite, Dolomite, Pyrite and kerogen) is plotted, along with P-impedance, S-impedance, bulk density, overburden stress, effective stress, and pore pressure for the well considered in this paper.

The rock formations in the study area are mainly composed of three major mineral groups: clay, silicates, and carbonates. The clay minerals are predominantly illite and chlorite, the silicates are mainly quartz, and the carbonates are chiefly calcite and dolomite. Additionally, the rock contains up to 6% pyrite by volume. The effective porosity of the rock varies significantly, ranging from 0% to 24%, and the kerogen content ranges from 0% to 15% by volume. Pore pressure measurements used in this study are derived from a regional pore pressure study conducted by [Loughry et al. in 2015](#).

Moreover, the study incorporates full azimuth Pressure wave (P-wave) data associated with P-wave to P-wave reflections (PP) seismic data. This seismic data has undergone extensive processing, including noise attenuation, multiple removal, and depth imaging. The final angle stacks around the well location, which were used for wavelet extraction. The frequency and phase of the extracted wavelets were found to be consistent, and this consistency was utilized for the PP AVA inversion to predict acoustic impedance and the V_p/V_s ratio.

This detailed analysis provides a robust framework for understanding the subsurface characteristics of the Midland Basin, offering valuable insights into the geological and petrophysical properties of the formations studied. Accurate petrophysical evaluation results, including volumes of chlorite, illite, bound water, quartz, calcite, dolomite, pyrite, and kerogen, are plotted along with P-impedance, S-impedance, bulk density, overburden stress, effective stress, and pore pressure for the well-considered in this paper.

Ternary Diagram

Various classification systems have been developed to categorize shales based on their mineralogical composition. These classifications help in understanding the behavior of the reservoir rock during drilling and production operations. Understanding the mineralogical composition is crucial for designing effective extraction techniques and predicting the reservoir's response to hydraulic fracturing. The ternary diagram

by Passey et al. (2010) from Exxon is a widely used tool for classifying organic mudstones based on their mineralogical composition. This diagram helps in understanding the distribution of key mineral components and their impact on reservoir quality.

SLB has developed a classification system for unconventional reservoirs, particularly organic mudstones, using a ternary diagram. This system, known as the sCore lithofacies classification, helps in understanding the mineralogical composition and quality of the reservoir. Here's a brief overview:

sCore Lithofacies Classification The sCore classification uses a ternary diagram with three apexes representing the primary mineral components: Quartz, Feldspar, and Mica (QFM), Carbonates (Calcite, Dolomite), Clays (Illite, Chlorite, Smectite/Illite). The entire Unconventional reservoir can be subdivided into 16 sub classes. The organic mudstone lithofacies log can be generated with minimal processing, and no interpretation is required. This facilitates consistency between multiple users which is important when comparing different organic shale plays. In addition, the lithofacies log can be generated from any kind of mineralogy data, it is not restricted to log data input. Figure. 2 shows an example of the specialized log display for sCore Lithofacies Classification.

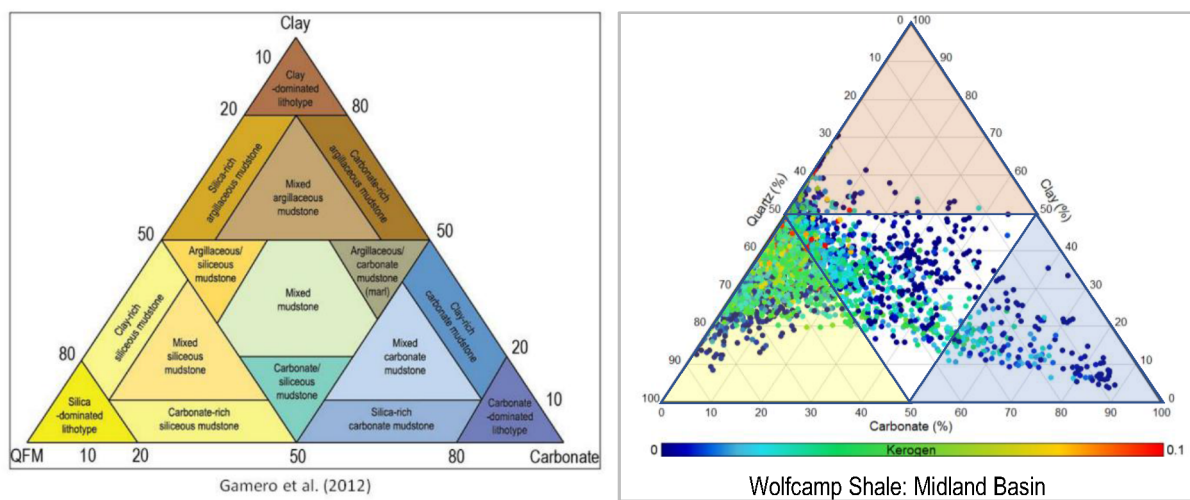


Figure 2a—LHS—The sCore ternary diagram displays 16 organic mudstone lithofacies (Gamero et al, 2012). The end members represent dry-weights of the main components of organic mudstone—clay (WCLA), carbonate (WCAR) and quartz, feldspar and mica (WQFM). Figure 2b: Ternary Diagram Wolfcamp Formation, understanding these constituents and their contributions is essential for optimizing drilling and completion strategies

Major Mineralogical Constituents of the Wolfcamp Formation and their Impact on Elastic Properties

Understanding these constituents and their contributions is essential for optimizing drilling and completion strategies in the Wolfcamp Formation.

1. **Silicates (Quartz, Feldspar)** High silica content, particularly quartz, provides brittleness and enhances the elastic properties of the rock. Both plagioclase and potassium feldspar are present.
2. **Carbonates (Calcite, Dolomite)** Carbonates like calcite and dolomite contribute to the rock's stiffness and overall elastic properties.
3. **Clays (Illite, Chlorite, Smectite/Illite)** Clays generally reduce the elastic modulus of the rock due to their ductile nature. However, the specific type and amount of clay can significantly influence the mechanical behavior of the formation.

Kerogen: The Wolfcamp Formation primarily contains Type II and Type III kerogen, which are indicative of oil and gas-prone organic matter. The kerogen content can vary, but it generally supports hydrocarbon generation.

Porosity (Organic and Inorganic Pore Space): The porosity of the Wolfcamp Formation can range widely, typically between 6% and 12%. This variation is due to the heterogeneous nature of the rock, which includes a mix of carbonates, shales, and silts.

Pore Pressure: The pore pressure in the Wolfcamp Formation is often overpressured, which means it exceeds the normal hydrostatic pressure. This overpressure can enhance hydrocarbon production but also requires careful management during drilling operations.

Due to the very low porosity, fluid properties and saturation levels do not significantly influence the elastic properties of the Wolfcamp Formation. The rock's mechanical behavior is more dependent on its mineralogical composition and the distribution of pore pressure.

Cross Plots

This below is the cross plot (Figure 3a) between porosity vs P-Velocity color coded based on carbonate (Red being Major carbonate mineral and blue min). This demonstrates the response of different minerals contribution on porosity. Low porosity is coming from high amount of carbonate. The Figure 3b Cross Plot is P-Velocity vs Carbonate: Velocity is increasing with increase in carbonate. There is an inflection point at a porosity of around 45%. So, the velocity is not changing significantly with the increase of carbonate till 45%. Hence, we see a positive increase of Velocity beyond it. This signifies carbonate as the main load bearing minerals where the carbonate is more than 45% of the volume and which means the sonic velocity is passing through this carbonate load bearing material.

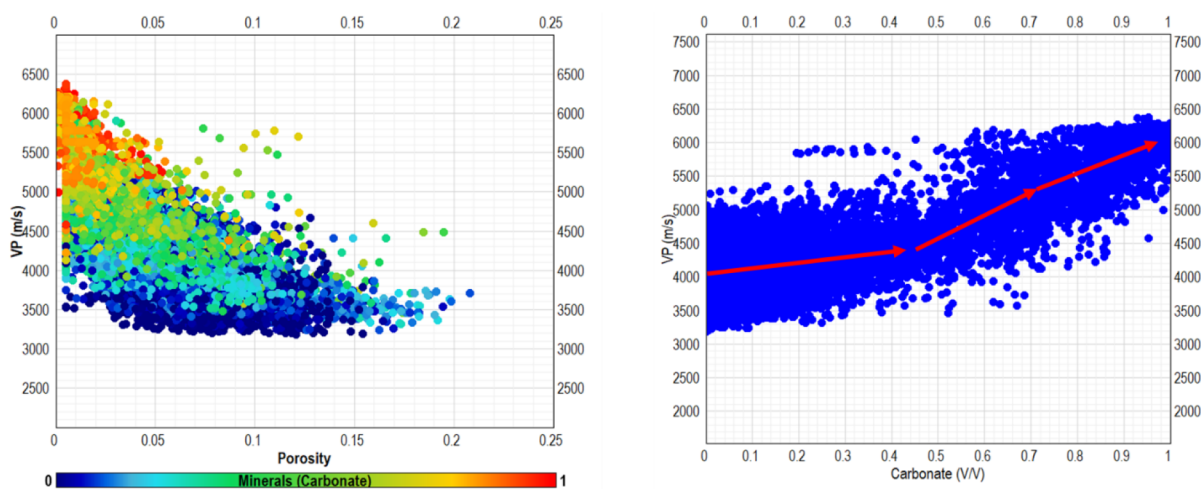


Figure 3a—Cross plot of porosity versus P-Velocity, color-coded based on carbonate content (red indicating a major carbonate mineral presence and blue indicating minimal carbonate). Figure 3b: The cross plot shows P-Velocity versus carbonate content.

The cross plots below (Figure 4a) is between porosity vs P-Velocity color coded based on Quartz (Red being Major Quartz mineral and blue minor). High porosity is contributed from high amount of quartz and low velocity.

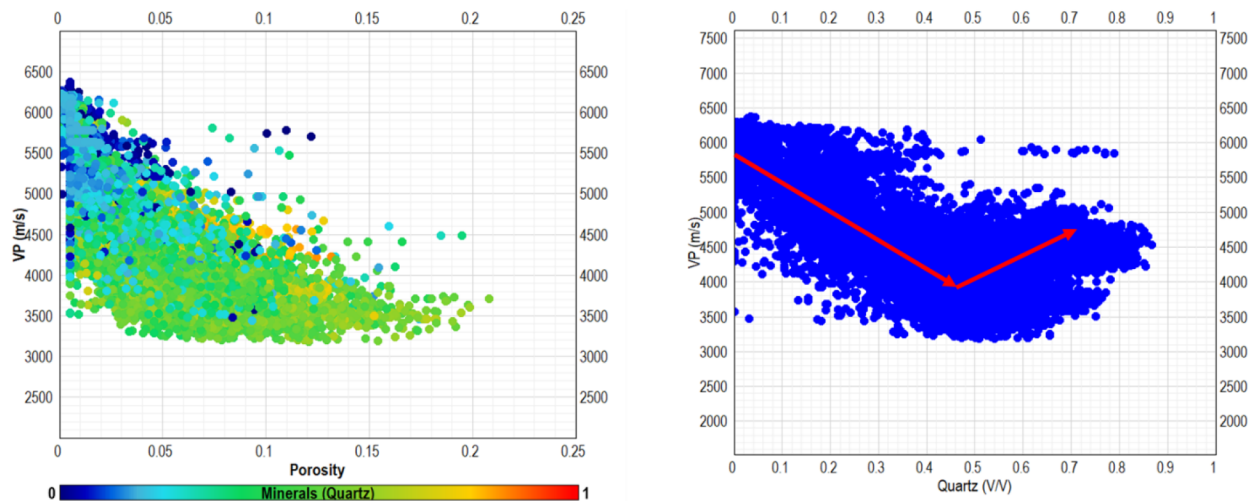


Figure 4a—The cross plots above show porosity versus P-Velocity, color-coded based on quartz content (red indicating major quartz mineral presence and blue indicating minor quartz presence). Figure 4b: The cross plot displays P-Velocity versus quartz content.

The Figure 4b Cross Plot is P-Velocity vs Quartz: There is an inflection point around 42%. When amount of quartz is very low (Calcitic Zones) the velocity is high mostly controlled by carbonate but when the amount of Quartz is above 42%, there is an increase of velocity with increase of Quartz and acting as the load bearing material

The below crossplot in Figure 5a and 5b is a typical signature of Clay. Clay is bringing everything down (Figure 5a). High clay zone has the lowest velocity for both Carbonate and Quartz, where the amount of clay is increasing, that is main load bearing material and bringing velocity down. 15% Porosity around 50% of clay.

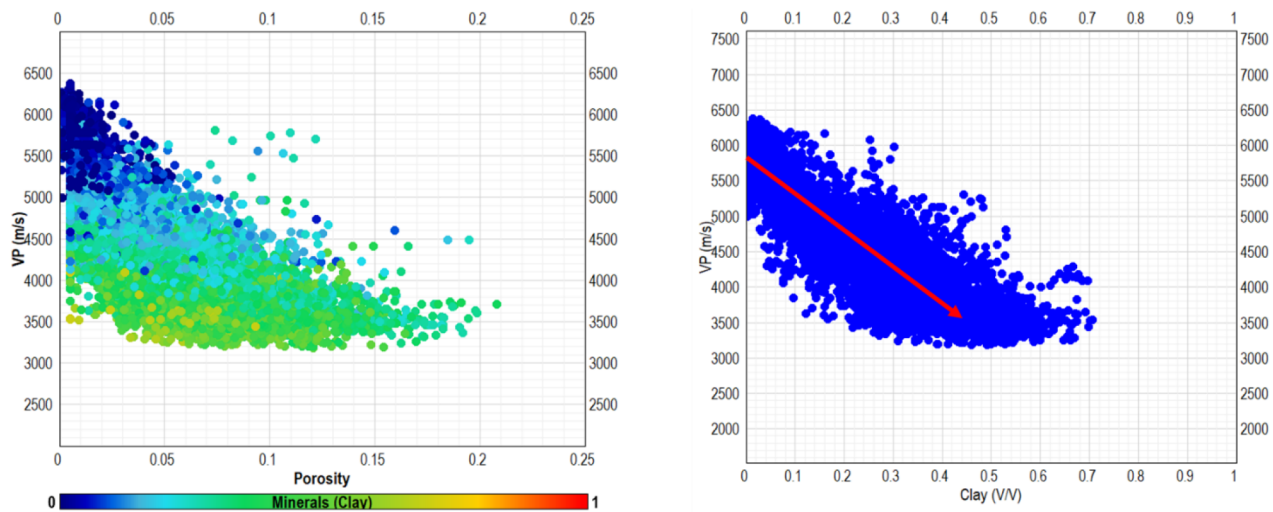


Figure 5a—The cross plots above show porosity versus P-Velocity, color-coded based on Clay content. Figure 5b: The cross plot displays P-Velocity versus clay content

The effect of Kerogen is plotted in Figure 6a and 6b. In case of Kerogen is low velocity and high porosity, does not act as load bearing material. It has complex relation with Porosity and pore pressure

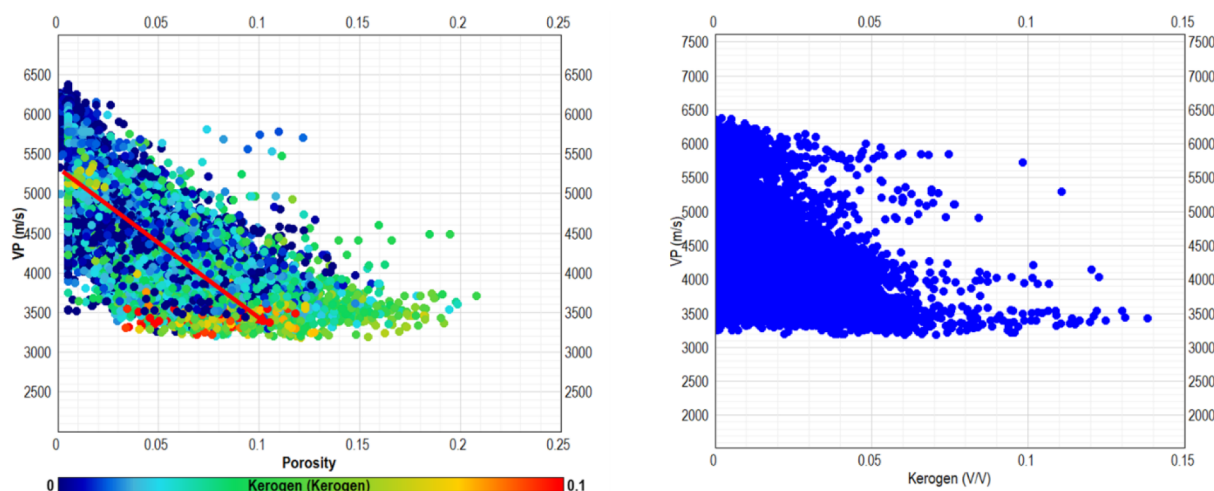


Figure 6a—The cross plots above show porosity versus P-Velocity, color-coded based on Kerogen. Figure 6b: The right cross plot displays P-Velocity versus Kerogen

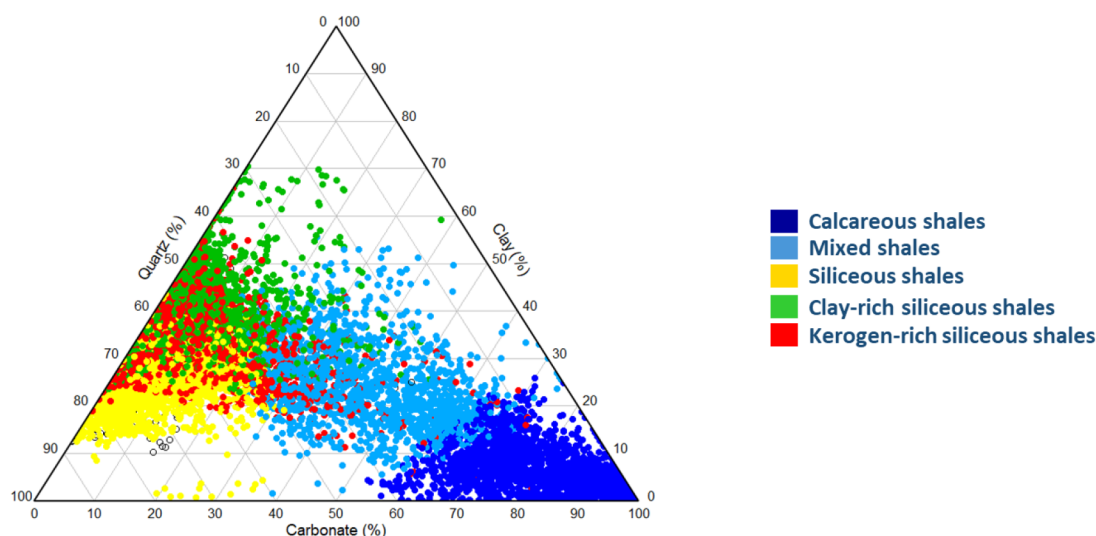


Figure 7—Mineralogy of the inorganic phase for a vertical well in the Wolfcamp Formation in the Delaware Basin, west Texas, represented by a ternary diagram. After the litho classification, we can see the distinctness of the 5 classes on the 5 parameters. There are box diagrams of the 5 classes with their range of parameters. In the Kerogen-rich siliceous shales we are able to see the highest amount of Kerogen. Similarly, we can see Quartz is max in Siliceous shales and Carbonate in Calcareous rocks. This provides some confidence in the classification.

Mineralogical composition, kerogen, and porosity exhibit significant influence on the characterization of elastic properties, such as the elastic moduli, bulk density, acoustic impedance and Vp/Vs ratio. Volume fractions of carbonate, clay, quartz, kerogen, and porosity from petrophysical interpretation were used to perform a facies classification of the reservoir. Principal component analysis was performed with these estimated volumes at the well location to reduce the dimension of the data and to find the orthogonal set of components required to explain the multidimensional space. K-means clustering was used to find the clusters with the principal components. Five classes were identified from the volume fractions:

1. Calcareous shales
2. Mixed shales
3. Siliceous shales
4. Clay-rich siliceous shales
5. Kerogen-rich siliceous shales

The principal components from PCA analysis are used in a K-means clustering algorithm to create the **HRA classification**.

Kmeans is an iterative algorithm that tries to partition the dataset into pre-defined distinct non-overlapping subgroups (clusters) where each data point belongs to only one group. The data points are assigned such that the sum of the squared distance between the data points and the cluster's centroid is at a minimum.

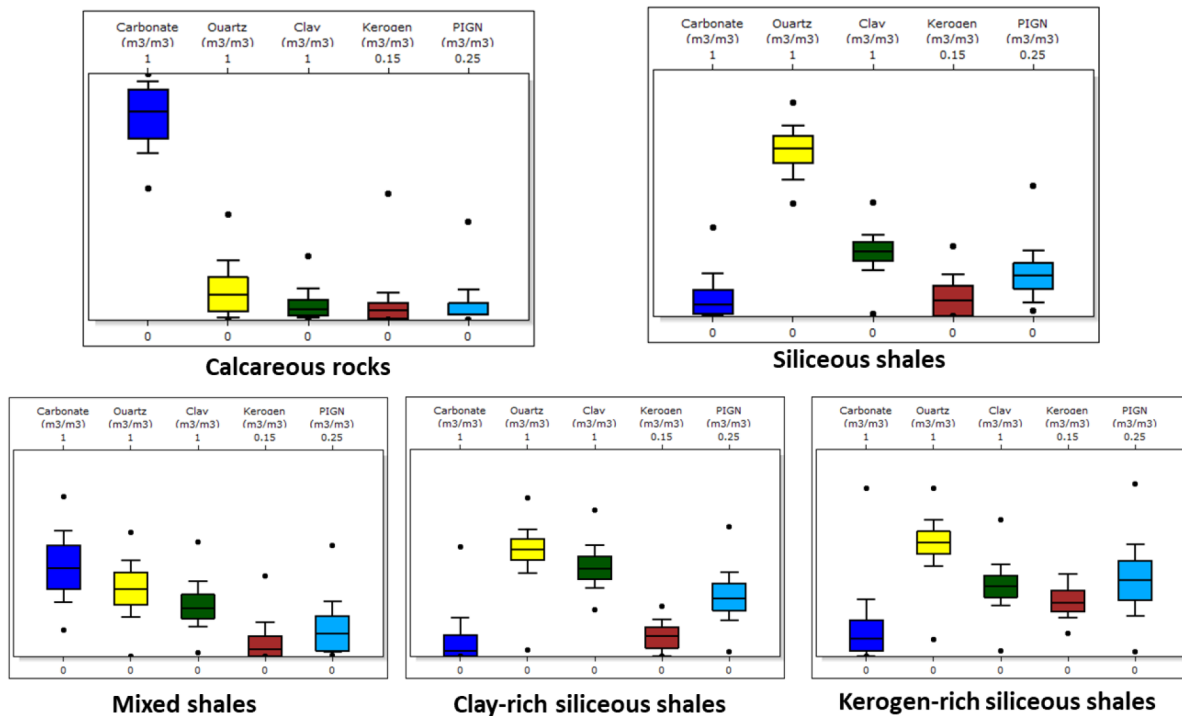


Figure 8—The box diagrams illustrate the range of parameters for each of the five classes based on the litho classification, the distinctness of the five classes across the five parameters is evident.

After the litho classification, we can see the distinctness of the 5 classes on the 5 parameters. There are box diagrams of the 5 classes with their range of parameters. In the Kerogen-rich siliceous shales we are able to see the highest amount of Kerogen. Similarly, we can see Quartz is max in Siliceous shales and Carbonate in Calcareous rocks. This provides some confidence in the classification.

Litho Classification and Rock Physics Model

The variation in lithology is depicted by showing several key parameters: the volume fraction of clays, quartz, carbonate, pyrite, kerogen, and effective porosity; vertical P-impedance; vertical S-impedance; bulk density; lithofacies (with kerogen-rich siliceous shales in red, clay-rich siliceous shales in green, siliceous shales in orange, mixed shale in cyan, and calcareous shale in blue); and pore pressure, overburden stress, and effective stress for the Clearfork to Atoka Formation sampled by the vertical well.

Mineralogical composition, kerogen, and porosity are crucial for characterizing elastic properties, such as the elastic moduli like acoustic impedance and the Vp/Vs ratio. Volume fractions of carbonate, clay, quartz, kerogen, and porosity derived from petrophysical interpretation were used to classify the reservoir facies. As you know, we obtain the Acoustic Impedance and Vp/Vs ratio from the prestack inversion. To link the litho classification derived from the well log with the seismic data, we need to establish a relationship between them.

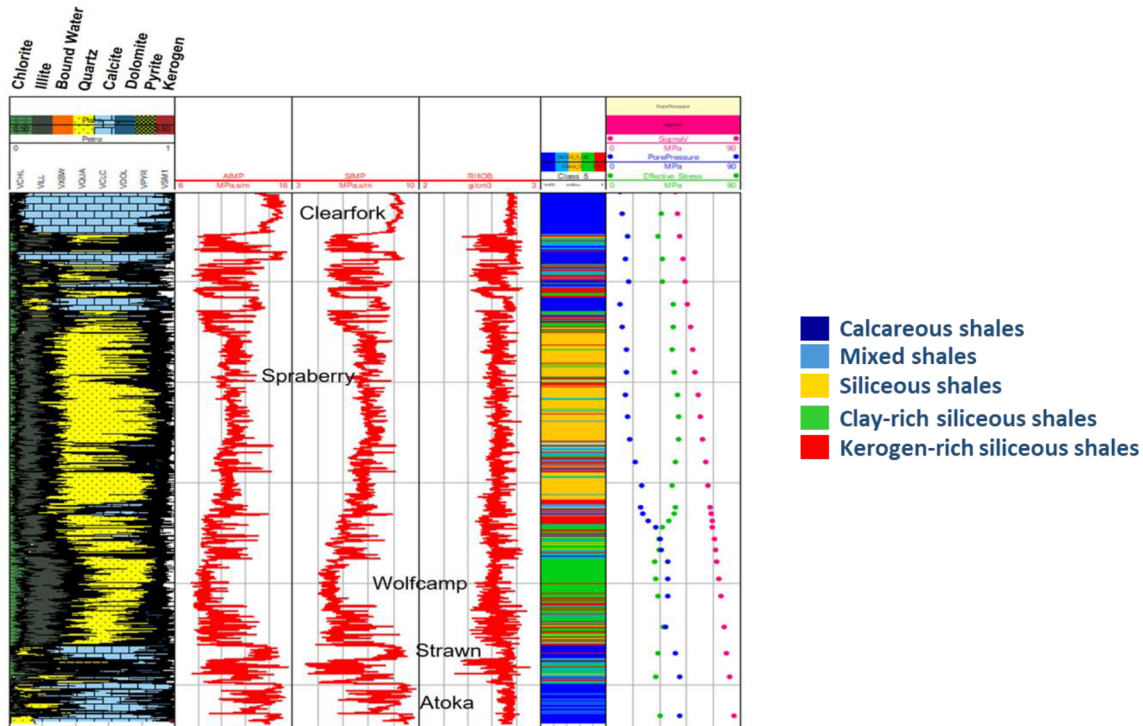


Figure 9—The log plot illustrates the variation in lithology, showcasing several key parameters. These include the volume fraction of clays, quartz, carbonate, pyrite, kerogen, and effective porosity. Additionally, it displays vertical P-impedance, vertical S-impedance, bulk density, and lithofacies. The lithofacies are color-coded as follows: kerogen-rich siliceous shales (red), clay-rich siliceous shales (green), siliceous shales (orange), mixed shale (cyan), and calcareous shale (blue). The figure also presents pore pressure, overburden stress, and effective stress for the Clearfork to Atoka Formation sampled by the vertical well.

We are now plotting the Acoustic Impedance versus the V_p/V_s ratio, along with the probability density function (PDF) of the five classes. The contours illustrate the probability distribution of each class within the dataset. The plot displays the general PDF with equal priors for the different classes in the Acoustic Impedance and V_p/V_s ratio plot.

Plot of equal prior representative PDF

Lithology estimation from elastic properties was done through Bayesian inverse problem formulation (Sengupta and Bachrach, 2007). Where, $(\mathbf{m}|\mathbf{F})$ is the likelihood of an elastic parameter, $\mathbf{m}$, given lithology, $\mathbf{F}$. and, $(\mathbf{F})$, is the prior probability density function (PDF) of each lithology. The lithoclasses are estimated by solving for the posterior probability function:

$$p(\mathbf{F}|\mathbf{m}) \sim L(\mathbf{m}|\mathbf{F})p(\mathbf{F})$$

Figure 10 shows the general PDF with equal priors for different classes in acoustic impedance and V_p/V_s plot.

The next step involves upscaling. Backus averaging is an effective harmonic averaging technique used to upscale sonic-log data to the seismic scale. This method involves harmonic averaging to determine the anisotropic elastic parameters that characterize seismic-wave propagation at low frequencies in a layered medium. In the plots, the black curves represent the well logs, while the red curves indicate the Backus average.

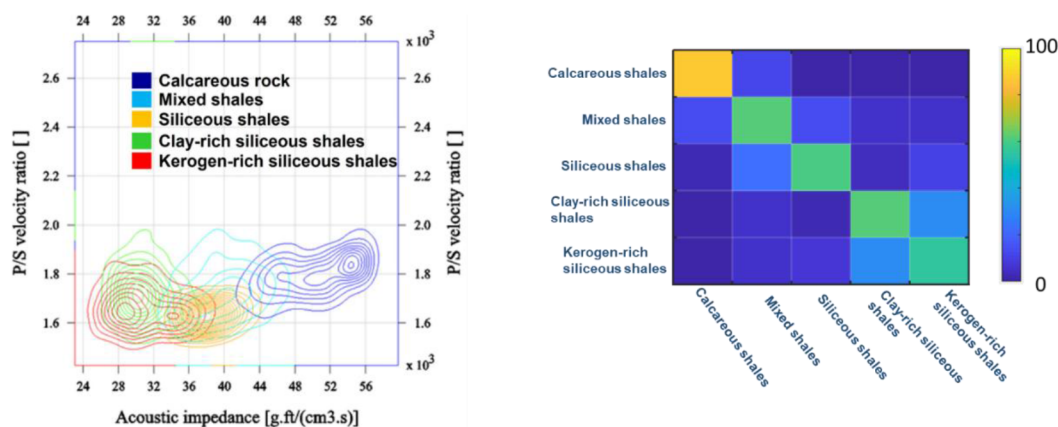


Figure 10—LHS Cross plot of Acoustic Impedance vs VPVS ratio with the probability density function of the 5 classes. The contours showed the probability distribution of the class varying in the dataset. RHS The plot shows the general PDF with equal priors for different classes in acoustic impedance and Vp/Vs plot.

A series of probability density functions (PDFs) were generated with variations in priors based on the stratigraphic distribution of lithoclasses. These PDFs utilize the Bayesian equation, where the prior probability describes the likelihood of a given value before considering new evidence. The four priors define the stratigraphy-based mineralogy understanding. This process was employed to reduce the overlap between different classes and improve the success rate for differentiating lithologies from prestack AVA inversion results. The stratigraphy-dependent prior-based PDFs were then applied to prestack AVA seismic inversion to generate lithological descriptions and associated probabilities for each class.

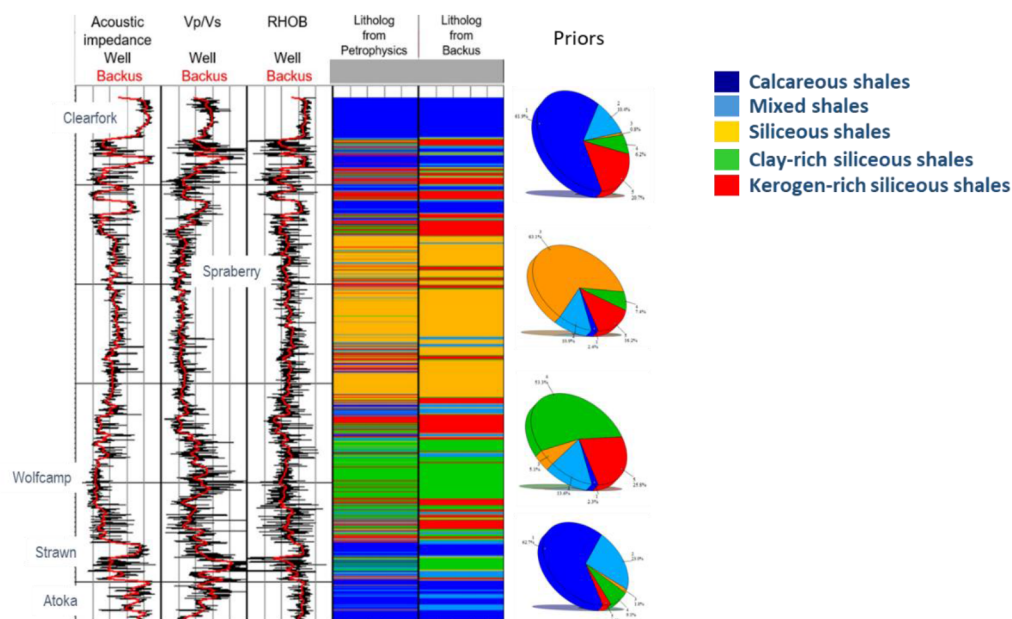


Figure 11—The upscaling of the logs using Backus averaging is depicted in red over the measured Acoustic Impedance, Vp/Vs, and Density. For comparison, the upscaled Litholog is also included in the plot. Subsequently, the stratigraphy-dependent prior-based PDFs were applied to the prestack AVA seismic inversion to generate lithological descriptions. The associated probabilities for each class are represented as pie charts.

Seismic Inversion Results and Predicted Litho-classes

The same series of stratigraphy dependent prior based PDFs were applied to prestack AVA seismic inversion to generate lithological description and associated probabilities for each class. This process was then used to reduce the overlap between the different classes and improve the statistics of success for differentiating

lithologies from prestack AVA inversion results. Now this plot is the Plot of orthogonal properties P-impedance and Vp/Vs (red, low; blue, high) obtained from prestack AVA inversion and the Backus upscaled logs are inserted at well location for comparison. This will show how accurately our seismic inversion has worked to reality from the logs (seismic data contains noises).

Probabilities Estimated from Lithology Classification

The lithoclass generates the probability of each class, considering the overlapping nature of the classes. Stratigraphically defined PDFs allow for the differentiation between the probability of siliceous shale and clay-rich siliceous shales. The previously described PDFs were utilized to generate upscaled lithoclasses from the Backus upscaled logs.

Figure 12 displays the vertical P-impedance and vertical Vp/Vs obtained from prestack AVA inversion for an inline section passing through the vertical well, as shown in Figure 1. The figure also presents the resulting litho-classification based on the maximum a posteriori estimate provided by the Bayesian analysis, using P-impedance and Vp/Vs derived from the previously described PDFs. Figure 13 illustrates the estimated probability associated with each class.

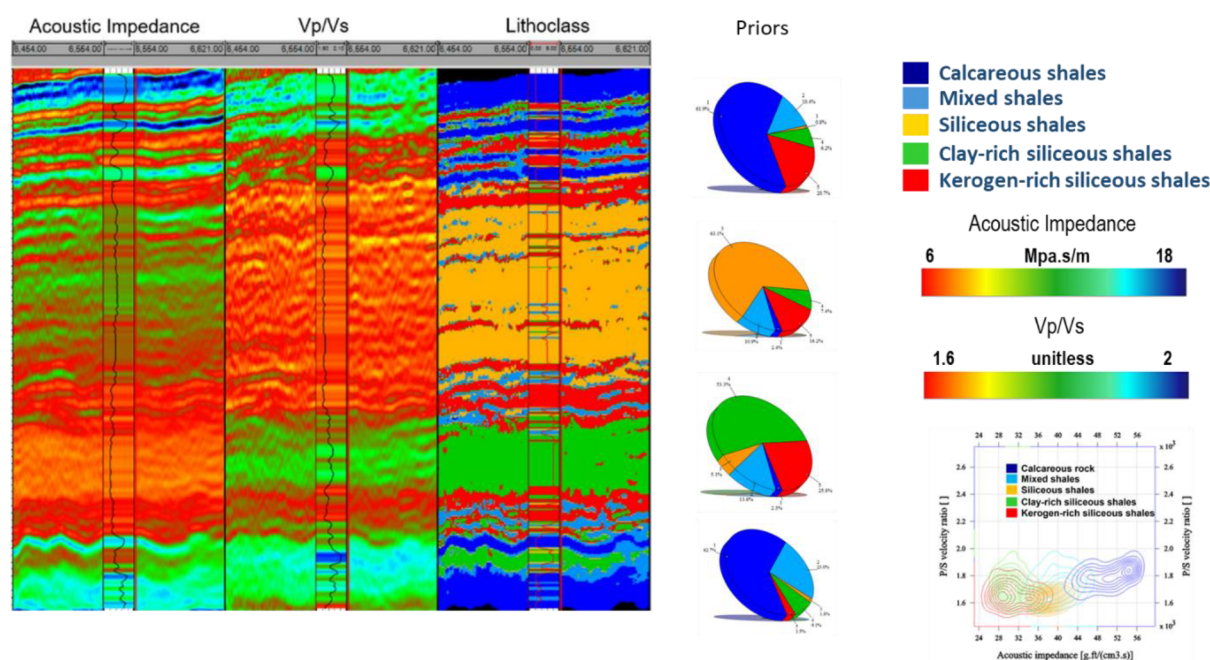


Figure 12—LHS plot presents a plot of P-impedance and Vp/Vs ratios, with red indicating low values and blue indicating high values, obtained from prestack AVA inversion for an inline section passing through the vertical well. The resulting lithoclass is also displayed, using the same color coding. For comparison, Backus upscaled logs are inserted at the well location. Stratigraphically variable priors are shown on the right side. The RHS cross Plot represents plot of equal prior representative PDF

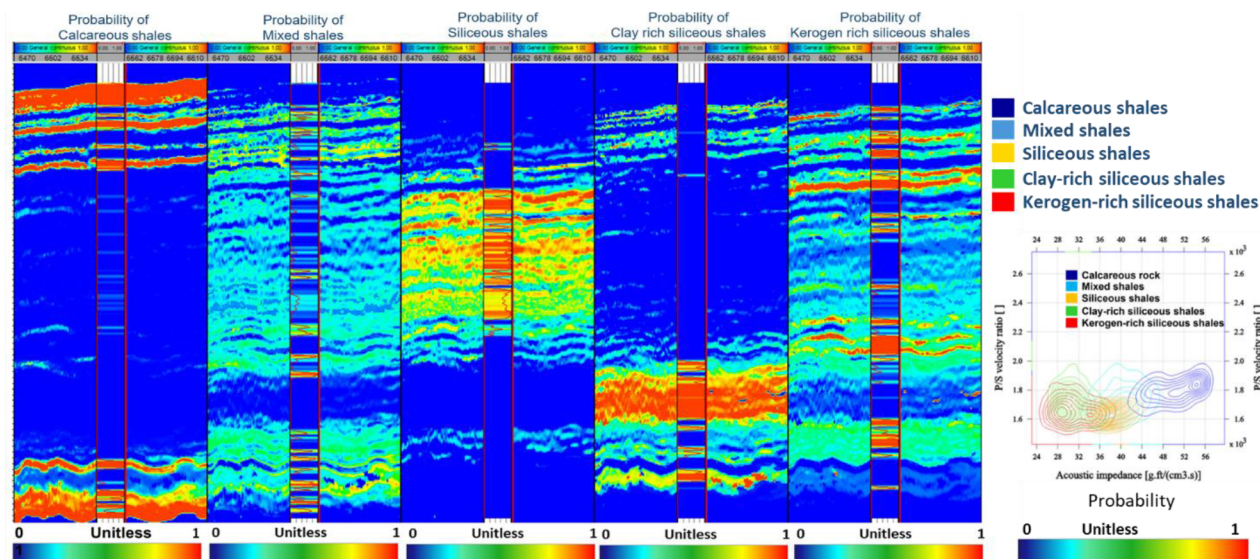


Figure 13—The plot illustrates a plot depicting the probabilities of various rock types. These include from LHS: (1) the probability of calcareous rock, (2) the probability of mixed shale, (3) the probability of siliceous shales, (4) the probability of clay-rich siliceous shale, and (5) the probability of kerogen-rich siliceous shale.

Using the rock physics model based on Inclusion-based Maxwell's theory, the final results were plotted. A Bayesian approach was employed to construct a stochastic model for predicting reservoir properties. This model predicted volumes of minerals, kerogen, porosity, and pore pressure from rock physics models using seismic inversion results. The plot illustrates the predicted reservoir properties against the measured logs at the well location and pore pressures from regional studies. The predicted results demonstrated a very high correlation with the measured data.

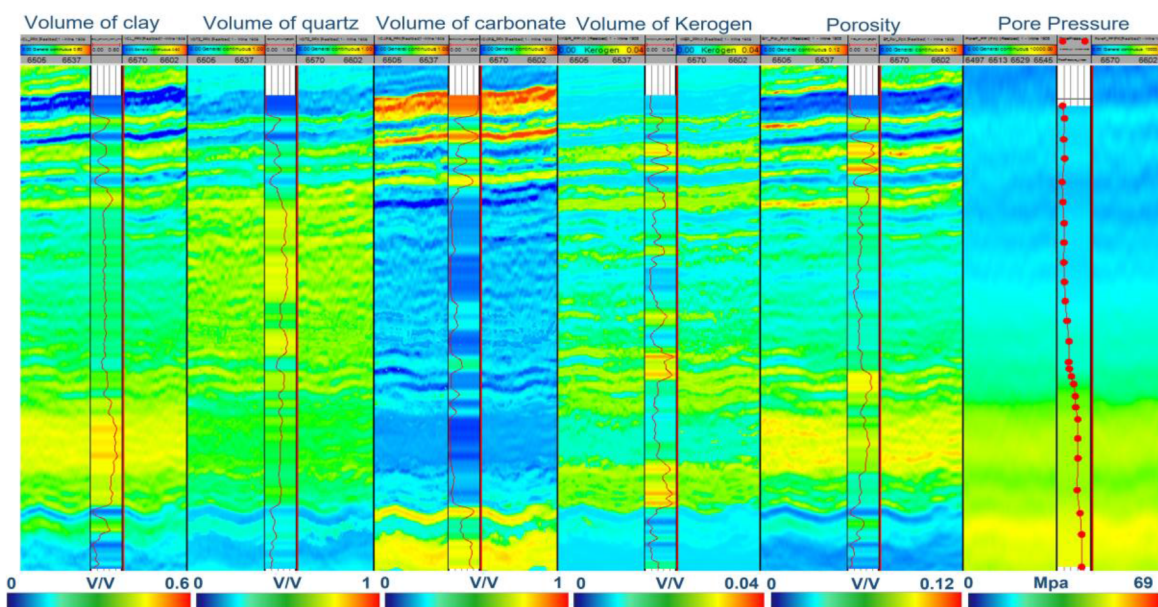


Figure 14—The plot presents the estimated reservoir properties derived from prestack AVA inversion for an inline section passing through the studied vertical well. The figure displays the following properties: (1) volume of clay, (2) volume of quartz, (3) volume of carbonate, (4) volume of kerogen, (5) porosity, and (6) pore pressure.

Conclusion

Predicting different facies across an unconventional field is crucial due to the heterogeneous nature of production from various types of source rock. To accurately map these facies, seismic inversion is essential. Well logs typically extend only a few inches from the borehole, so to understand spatial variations, it is necessary to propagate well properties throughout the entire seismic cube.

In this study, inverted values of P-impedance and V_p/V_s ratio obtained through prestack AVA inversion in the Midland Basin, West Texas, were used. A Bayesian method was employed to predict the dominant lithologies using novel stratigraphy-dependent prior-based PDFs. These predictions were consistent with lithologies identified through petrophysical analysis at a specific well location. The primary drilling interest was in the kerogen-rich siliceous shale of the Wolfcamp section. The distribution and associated probability of this shale provided critical information for the landing and drilling of lateral wells. Notably, the kerogen-rich siliceous shale exhibited the highest effective porosity.

It has been demonstrated that properties such as volume of kerogen and porosity can be estimated by integrating the rock-physics model with seismic inversion results for each facies volume. While pore pressure remains central to rock property estimation, this study presents a method to estimate pore pressure along with other reservoir properties using seismic data as the starting point.

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